Element H: Prototype Testing And Data Collection Plan

	Requirement	Priority	Testing Setup And Procedure	Test Cases And # Of Trials	Pass/Fail Criteria And Goal
1	Structure - There must be two rooms, one 12"x12" (room 1), and one 6"x12" (room 2)	High	1) Use a measuring Tool on the Outer Wall shared by both rooms and make sure it is 18” by 12” 2)Use a measuring tool in room 1 and measure the length of the room, make it 12”, and make sure the wall is 12” tall 3) Repeat Step 2 in room 2, but make sure this time it is 6” by 12”	1) Measure each wall on both ends to ensure equality. (12 trials at least)	**Pass** - All sides have the correct dimension with a .5” room for error and are symmetrical on both sides. **Fail** - One or more of the dimensions are off by more than .5”. **Goal -** All dimensions are right on target.
2	Removable roof	Me…	1)Use a Dial caliper to make sure the corner gaps are not too big 2)Mildly shake the model home with the roof to ensure it stays on 3)Lean the model slightly and ensure the roof stays one. 4)Test the removal mechanism, ensure it does not take long	1) Lean the model in all directions (4 trials) 2)Shake the Model(3 trials)	**Pass -** The Roof is stable but is easily removed and reattached. **Fail -** The Roof comes off easily without any disturbance. The roof does not come off easily and can be easily reattached. **Goal -**The model can keep the roof on when it experiences a disturbance, and come off when needed, and can be easily put back
3	*Openings* *One doorway* 6.5" x 2.5" leading out of the	High	1) Measure all the openings and make sure they have the same dimensions 2) Use Calipers to make sure that the door closes fully and	Use a measuring tool to check dimensions, and use a caliper to see gaps	**Pass** - All openings have the correct dimension with a .25” room for error and are symmetrical on both sides.

	front room 1 outside, and one doorway of the same size connecting both rooms. One window on two exterior walls for the larger room (two windows).		no gaps	**Trials** - As many as needed till it is right	**Fail** - Dimensions are off by more than .25", and there are gaps in the openings when closed that will allow oxygen in. **Goal** - All dimensions for the openings are perfect, and no gaps exist in the openings when closed/
4	Furniture The furniture must be representative of real furniture, i.e., materials selected for furniture must be similar to materials used in actual homes and incorporate materials such as foam plastic, cotton, wood, and cloth.	High	1) Ensure there are two chairs with the dimensions 2.5 x 2.5 x 2, ensure they are made using crafting sticks 2) Ensure there are two desks with dimensions 4 x 2 x 3 using crafting sticks 3) Ensure the bed is made using Foam, crafting sticks with the dimensions 6 x 3 x 2. Wrap the foam in fabric	1)All items are made of the correct material and use 2)Use any measuring tool to get the right dimensions. **Trials** - As many as needed till it is right	**Pass -** The furniture has the right dimensions, and the correct materials are used in all situations. **Fail -** The dimensions are off, and some materials not listed here are used. **Goal -** The furniture has the correct dimensions, and the right materials are used in all situations.
5	Wall and Floor Finishings - Carpet for all the rooms - Curtains for windows - One 3" by 3" poster	Low	1) Ensure the carpet covers all parts of the model and is installed securely. 2) Ensure Curtains Cover all openings 3) Ensure the poster is on the wall	Use a caliper to measure the size of the poster **Trial**- As many as needed till it is right	**Pass -** All furnishings are installed **Fail -** None or all furnishings are not added **Goal -** All furnishings are installed and fairly aesthetic
6	Cost Total cost must not exceed $50	Me…	1) Make a budget Plan 2) Document all expenditures	Use a Google Doc to do the documentation of all expenditures	**Pass** - The total cost is less than $50 **Fail -** Total cost is over $50 **Goal -** Saving as much money as possible without compromising quality

7	Time for suppression Fire must be suppressed in less than 5 minutes.	High	1) Set a timer 2) Activate the suppression system 3) Test suppression in an area of similar size to the model Home with similar flammable materials 4) Document the time after each step till below 5 minutes	1) The test must be done in a safe environment 2) The test must simulate the actual conditions that the system will be put in real situations **Trials**: At least seven to ensure it works consistently, or until we get it under 5 minutes	**Pass -** Fire is suppressed under 5 minutes **Fail -** The system takes longer than five minutes to suppress the fire **Goal -** The suppression system activates simultaneously with the notification system and suppresses the fire in under 2 minutes
8	Detection And Notification Time Fire must be detected, and notification must start in under 30 seconds	High	1) Set a timer 2) Read the serial monitor for reading from our sensors(MQ2 gas sensors and IR sensors) 3) When the gas Sensor has a ppm of 100 and the IR sensor is activated 4) The notification system should activate 5) Record how long this takes to happen till under 30 seconds	Perform the test in a safe environment and ensure there is a fire and Smoke to perform the test **Trials**: At least seven to ensure it works consistently, or until we get it under 30 seconds	**Pass** - The system is activated in less than 30 seconds **Fail -** The Detection and The Notification System takes more than 30 seconds to activate. **Goal -** The detection and Notification system activates in significantly less than 30 seconds